

Homework 2

Decision Variables

x_1 = Number of hand bags.

x_2 = Number of shoulder bags.

Objective Function

$$\text{MAX } 5.5x_1 + 1x_2$$

Constraints

$$5x_1 + x_2 \leq 200$$

- machine minute limit.

$$2.5x_1 + 4x_2 \geq 125$$

- inches of recycled leather.

$$3x_2 \leq 90 + 2x_1$$

- fabric stock

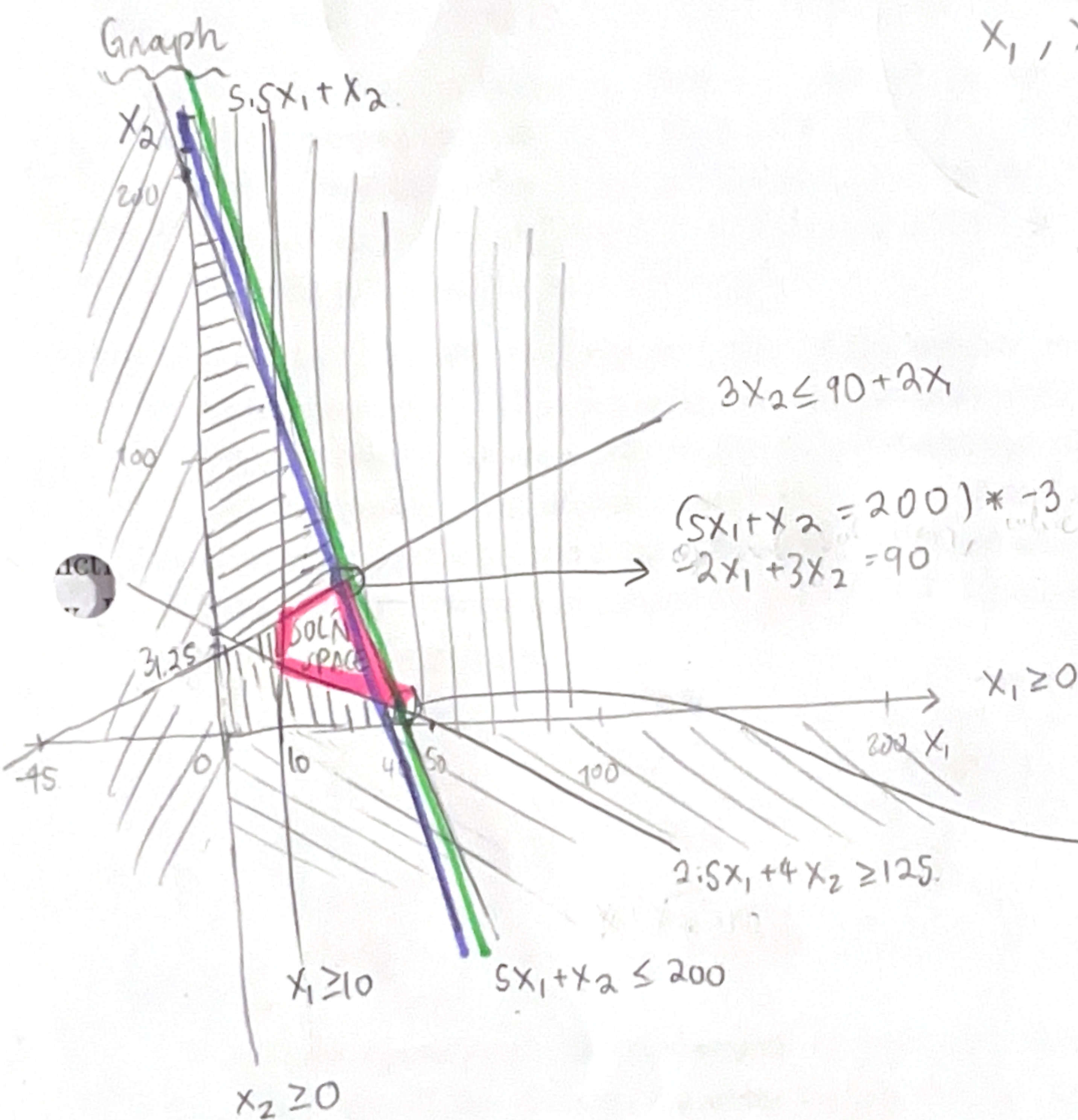
$$x_1 \geq 10$$

- retailer expectation

$$x_1, x_2 \geq 0$$

- non negative

Graph



$$\begin{aligned} (5x_1 + x_2 = 200) * -3 &\Rightarrow -15x_1 - 3x_2 = -600 \\ -2x_1 + 3x_2 &= 90 \\ \hline -17x_1 &= -510 \end{aligned}$$

$$x_1 = 30$$

$$x_2 = 50$$

$$\star 5.5x_1 + x_2 = 215$$

$$\begin{aligned} 2.5x_1 + 4x_2 &= 125 \\ (5x_1 + x_2 = 200) * -4 &\Rightarrow -20x_1 - 4x_2 = -800 \\ \hline -17.5x_1 &= -675 \end{aligned}$$

$$\Rightarrow x_1 = 38.571, x_2 = 7.143$$

$$\star 5.5x_1 + x_2 = \boxed{219.29}$$

is the optimal solution.